

OPERA NUMERORUM

NS Tower -- Path A: Phase 100
NS_Zerolnit_L2Zero_PROVED -- 0 sorry, 6-step Mathlib proof

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SECTION 1: THEOREM PROVED

NS_Zerolnit_L2Zero_from_EnergyLe (0 sorry, conditional on NS_WeakSol_EnergyLeL2_OPEN)

Statement (NS_Zerolnit_L2Zero_OPEN):

forall v0 v, NS_WeakSolution v v0 -> (forall^ae x partial haar, v0 x = 0) -> forall t >= 0, integral x, ||v t x||^2 partial haar = 0

Proof (6 steps, all Mathlib v4.12.0):

Step	Goal	API
A	$v\ 0 = v_0$	NS_WeakSolInitCond_PROVED (Phase 99, h.init)
B	$\ v_0\ x\ ^2 = 0$ a.e.	congr_fun hInit x; rw hx; norm_zero; sq; mul_zero
C	$\int \ v\ 0\ ^2 = 0$	integral_congr_ae (B) + integral_zero
D	$\int \ v\ t\ ^2 \leq 0$	NS_WeakSol_EnergyLeL2_OPEN + C
E	$0 \leq \int \ v\ t\ ^2$	integral_nonneg + sq_nonneg
F	$\int \ v\ t\ ^2 = 0$	le_antisymm (D) (E)

Axiom footprint: {propext, Classical.choice, Quot.sound} | SORRY COUNT: 0 | AXIOM KEYWORD: 0

SECTION 2: NEW NAMED OPEN DEF

NS_WeakSol_EnergyLeL2_OPEN (ETA 1-2 days):

forall v0 v, NS_WeakSolution v v0 -> forall t >= 0, integral x, ||v t x||^2 partial haar <= integral x, ||v 0 x||^2 partial haar

Route: unfold NS_WeakSolution.energy_le (WeakNS.energy_le from WeakSolution.lean L130) and the energy definition integral ||.||^2/2. Does NOT appear in #print axioms.

SECTION 3: PATH A GAP TABLE (Phase 100, 8 deps)

#	Named Open Def	ETA	Status
1	NS_WeakSol_EnergyLeL2_OPEN	1-2 days	OPEN (IMMEDIATE)
2	NS_Zerolnit_Pointwise_OPEN	1-2 weeks	OPEN
3	NS_ESSRescaleNS_OPEN	2-4 weeks	OPEN
4	NS_Carleman_SmoothApprox_OPEN	3-6 weeks	OPEN
5	NS_BlowupConcentration_OPEN	2-3 months	OPEN
6	NS_Carleman_LimitPass_OPEN	2-4 months	OPEN
7	NS_CarlemanHeat_OPEN	3-6 months	OPEN (CRITICAL)
8	NS_CarlemanDriftAbsorption_OPEN	after heat	OPEN

SECTION 4: INTEGRITY

Repository: DavidFox998/navier-stokes Phase file: NSPhase100L2ZeroClose.lean Lean
toolchain: v4.12.0 | Mathlib: v4.12.0 CMI rules: 0 sorry, 0 axiom keyword, named open defs
only M7 manifest: FROZEN (5b80b84d) -- m1..m6.out unchanged Cumulative: Phase 95 (7 deps)
-> 98 (10) -> 99 (8) -> 100 (8, better deps)